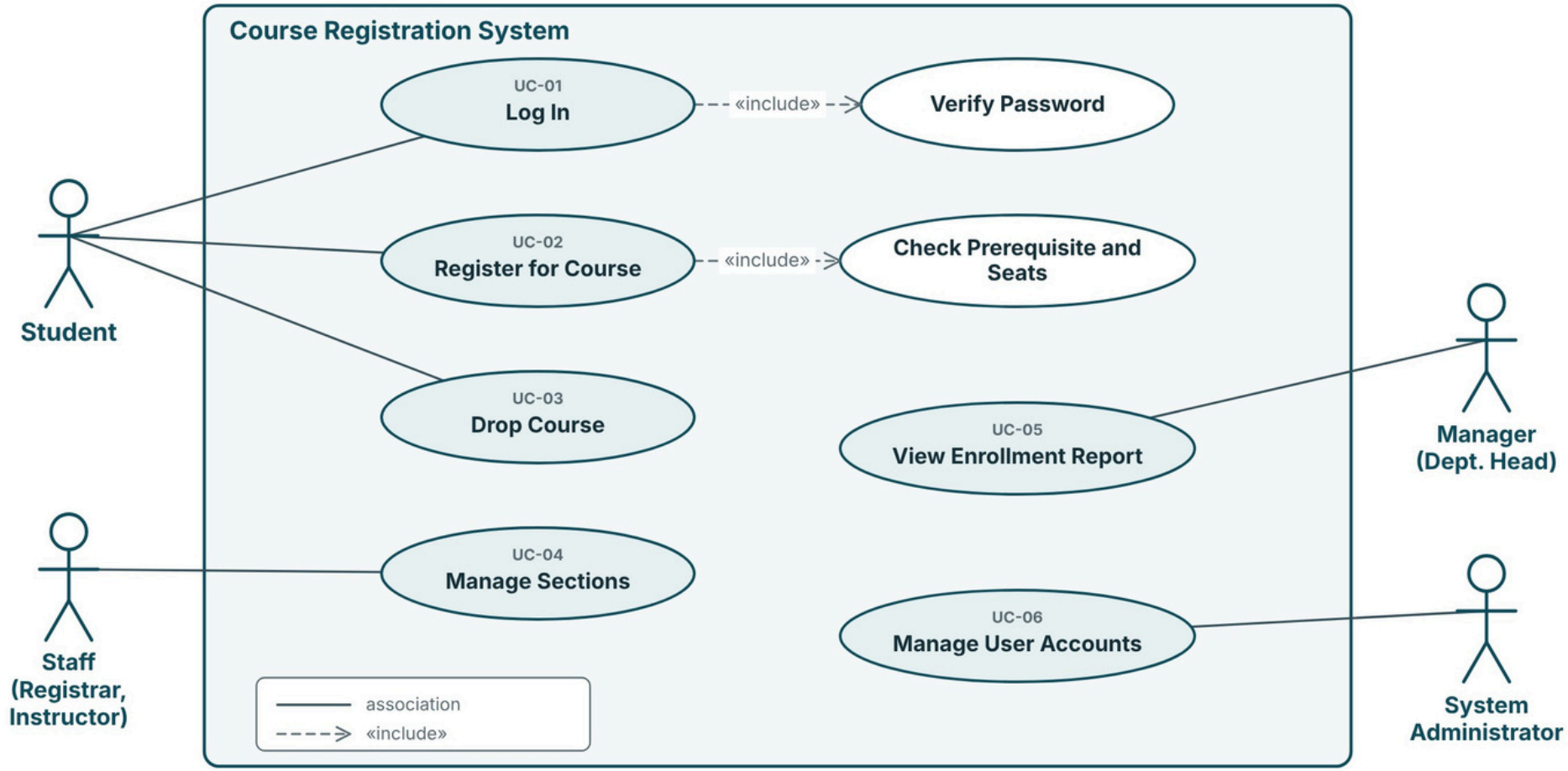
Use Case Diagram: Course Registration System



Explanation Four actors use six use cases. The Student logs in (including password verification), registers for a course (including a check of the prerequisite and seats) and drops a course. Staff manage sections, the Manager views enrollment reports and the System Administrator manages user accounts.

Use Case Specifications (UC-02 to UC-05)

UC-02 Register for Course — Student

Precondition: The student is logged in and the registration period is open.

Main flow: 1. The student searches and selects a section. 2. The system checks the prerequisite. 3. The system checks that a seat is free. 4. The system checks the time conflict. 5. The system creates the registration, adds 1 to seats_taken and shows a confirmation.

Alternative: Prerequisite missing or time conflict: the request is rejected with the reason. Section full: the system offers the waitlist. **Postcondition:** A REGISTRATION record exists with status Registered.

UC-03 Drop Course — Student

Precondition: The student has a registration with status Registered or Waitlisted.

Main flow: 1. The student selects the course and confirms the drop. 2. The system checks the drop deadline. 3. The system sets the status to Dropped and releases the seat. 4. The system promotes the first waitlisted student, if any. 5. The system shows a confirmation.

Alternative: Deadline passed: the drop is refused and the student is told why.

Postcondition: The registration status is Dropped and the seat is free again.

UC-04 Manage Sections

Precondition: The staff member is logged in.

Main flow: 1. Staff open the Manage Sections screen. 2. Staff create or edit a section: course, day and time, room, capacity and instructor. 3. The system checks room and instructor conflicts. 4. The system saves the section.

Alternative: Conflict found: the system shows a message and staff change the time or room.

Postcondition: The SECTION record is saved and students can register for it.

UC-05 View Enrollment Report

Precondition: The manager is logged in.

Main flow: 1. The manager opens the report page and selects a semester. 2. The system reads capacity and seats_taken of every section. 3. The system shows seats used, full sections and waitlisted students.

Alternative: No data for the semester: the system shows an empty report message.

Postcondition: The report is displayed; no data is changed.

Explanation Each specification lists the actor, the conditions before and after, the main step-by-step flow and what happens when a check fails. UC-02 and UC-03 follow business rules BR-1 to BR-4; UC-04 and UC-05 serve staff and managers.